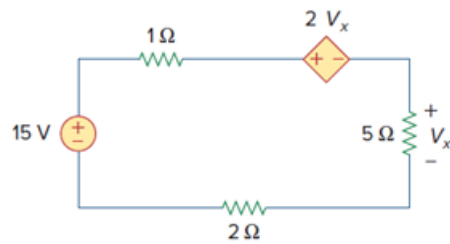


Problem

Find V_x in the circuit of Fig. 2.85.

Figure 2.85



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

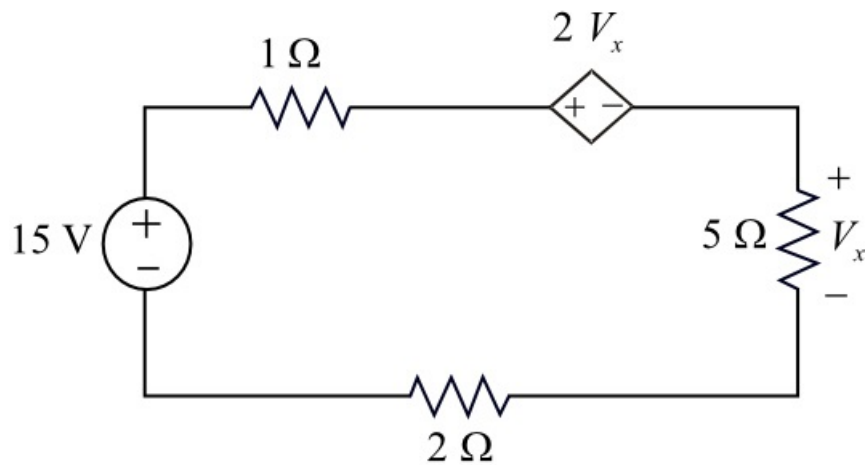


Figure 1

Step 2 of 4

Apply Kirchhoff voltage law around the loop 1 for the shown below circuit.

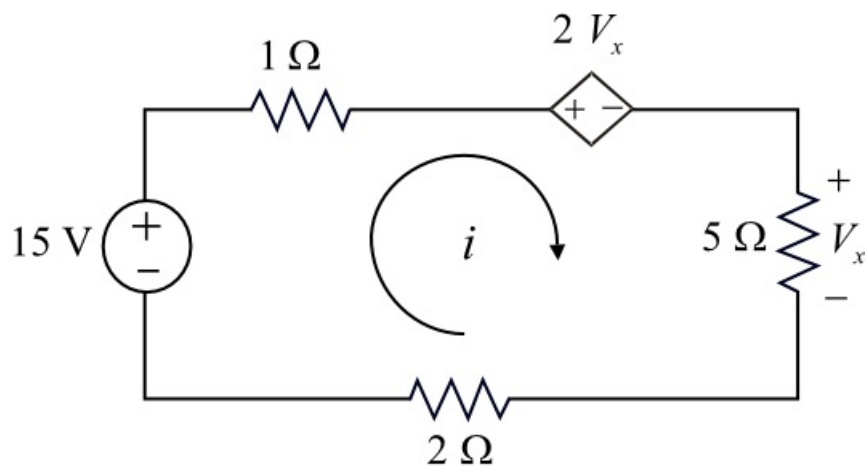


Figure 2

Step 3 of 4

Apply Kirchhoff voltage law around the loop 1.

$$-15 + i + 2V_x + 5i + 2i = 0 \dots\dots (1)$$

Apply ohm's law to the $5\ \Omega$ resistor.

$$V_x = 5i \dots\dots (2)$$

Substitute $5i$ for V_x in equation (1).

$$-15 + i + 2(5i) + 5i + 2i = 0$$

Rewrite the above equation.

$$18i = 15$$

$$i = \frac{15}{18}$$

$$= 0.833\text{ A}$$

Step 4 of 4

Substitute 0.833 A for i in equation (2).

$$V_x = 5(0.833)$$

$$= 4.16\text{ V}$$

Therefore the voltage V_x across the $5\ \Omega$ resistor is 4.16 V .